

DATA STRUCTURES

LAB PROGRAMS

19.Heap sort

```
#include <stdio.h>

void heapify(int arr[], int n, int i) {

    int largest = i, left = 2*i+1, right = 2*i+2;

    if(left < n && arr[left] > arr[largest])

        largest = left;

    if(right < n && arr[right] > arr[largest])

        largest = right;

    if(largest != i) {

        int temp = arr[i];

        arr[i] = arr[largest];

        arr[largest] = temp;

        heapify(arr, n, largest);

    }

}

void heapSort(int arr[], int n) {

    for(int i = n/2-1; i >= 0; i--)

        heapify(arr, n, i);

    for(int i = n-1; i > 0; i--) {

        int temp = arr[0];

        arr[0] = arr[i];
```

```
        arr[i] = temp;
        heapify(arr, i, 0);
    }
}

int main() {
    int arr[] = {23,56,12,34,78,29};
    int n = 6;
    heapSort(arr,n);
    printf("Sorted Array:\n");
    for(int i=0;i<n;i++)
        printf("%d ",arr[i]);
    return 0;
}
```

Output

Sorted Array:

12 23 29 34 56 78

=== Code Execution Successful ===